

BTLO Challenge Report

Meta — Digital Forensics

Analyst	Abiral Timalina
Platform	Blue Team Labs Online (BTLO)
Category	Digital Forensics
Difficulty	Easy
Points	10
Tools Used	Kali Linux, ExifTool, Google Images (reverse image search)

1. Scenario

The attached images were posted by a criminal on the run, with the caption "I'm roaming free. You will never catch me." The objective is to analyze the embedded metadata to identify the camera used, when the photo was taken, any hidden comments, and the criminal's likely location.

2. Methodology

2.1 File Acquisition

The challenge archive was password-protected using AES encryption, which the default unzip utility on Kali could not handle. The archive was extracted using 7-Zip instead:

```
7z x <archive>.zip
```

This produced two files: `uploaded_1.JPG` and `uploaded_2.png`.

2.2 Metadata Extraction — Image 1 (JPG)

ExifTool was used to extract the full metadata set from the first image:

```
exiftool uploaded_1.JPG
```

The output revealed the camera make/model, original capture timestamp, and a `Comment` field embedded in the EXIF data reading: *"relying on altered metadata to catch me?"*

The GPS coordinates in this same file were checked for validity. The recorded longitude (279° W) exceeds the maximum possible real-world value of 180°, confirming the GPS data had been deliberately falsified — consistent with the embedded taunt in the comment field. This GPS data was therefore excluded from further analysis.

2.3 Metadata Extraction — Image 2 (PNG)

```
exiftool uploaded_2.png
```

PNG files do not carry EXIF metadata the way JPEGs do, and the extraction confirmed no GPS or camera data was present. As the metadata trail was a dead end for this file, the image was opened directly for visual inspection instead.

2.4 Visual / OSINT Analysis

The second image showed multi-tiered pagoda-style architecture consistent with traditional Newari construction. A reverse image search (Google Images) was performed to confirm the exact landmark, returning a match for **Basantapur Palace**, part of the UNESCO World Heritage site of **Kathmandu Durbar Square**.

3. Findings

#	Question	Answer
1	What is the camera model?	Canon EOS 550D
2	When was the picture taken?	2021:11:02 13:20:23
3	What does the comment on the first image say?	relying on altered metadata to catch me?
4	Where could the criminal be?	Kathmandu

4. Lessons Learned

- Metadata should not be trusted blindly — validating GPS coordinate ranges (latitude $\pm 90^\circ$, longitude $\pm 180^\circ$) is a quick and effective way to detect tampered EXIF data.
- When one evidence source is deliberately falsified, it is often a deliberate distraction from a second, untampered source — in this case, the visual content of a second image with no embedded metadata at all.
- Reverse image search remains a fast, reliable OSINT technique for geolocating landmarks when metadata is unavailable or unreliable.